Tasnim Amara, B.A.SC. (2030)

tasnimamara | Tasnim Amara | Portfolio | tamara@uwaterloo.ca | +01.226.400.2623

Skills

Programming	Proficient in Python, Java, MATLAB, HTML/CSS
Engineering Software	Skilled with using Microsoft Office, SIMULINK, SolidWorks, Altium Designer, Traxmaker PRO, Isaac Sim
Hardware/Laboratory	Well-versed in Micropipetting, Electroplating, Material Performance Testing, Literature Review, PIC Microcontrollers, Arduino, PCB Assembly & Troubleshooting
Soft Skills	Experience with Project Coordination, Technical Documentation, Team Collaboration

Work Experience

Undergraduate Research Assistant — **Dec 2025 - Present**
Multi-Scale Additive Manufacturing Lab, Kitchener, ON

- Collaborating with PhD researchers to bridge academic materials science with industrial applications, contributing to sensor reliability assessments and biomedical device innovation.
- Fabricating precision sensor components via electroplating and thin-film deposition, operating Astrojet systems to achieve micron-scale accuracy for industrial structural health monitoring.
- Developing thermally stable composite formulations through systematic material testing and data collection under simulated high-stress environments.
- Conducting comprehensive literature reviews on 3D printing technologies for biocompatible silicone, analyzing 20+ suppliers and presenting findings to stakeholder companies.
- Evaluating printer specifications and material certifications to recommend cost-effective, production-ready solutions for customizable medical-grade hearing aid manufacturing.

Summer Reading Club Facilitator — **Jun 2025 - Aug 2025**
Waterloo Public Library, Waterloo, ON

- Acted as the welcoming face for families, managing registrations and inquiries, directly increasing program retention through personalized interactions.
- Collaborated with a team to plan and execute a diverse schedule of 4 weekly activities, demonstrating strong project coordination and time-management skills.
- Coordinated a large-scale end-of-summer party for over 200 attendees, improving logistical planning and teamwork abilities.

Projects

Webcam-Controlled Robot — *Arduino, MATLAB/SIMULINK, Computer Vision*
Developing a webcam-controlled rover for path following, object manipulation (forklift attachment), and obstacle avoidance. Programming Arduino Nano 33 IoT for real-time motor control and integrating computer vision algorithms in MATLAB for visual navigation. Building mechanical structure, integrating servo motors/drivers, and troubleshooting hardware/circuit implementation.

Firefighter Flame Extinguisher Robot — *Microcontrollers, PCB Design, Troubleshooting*
Built an autonomous fire-extinguishing robot that detects flames using IR/ultrasonic sensors and navigates to the source. Designed custom PCBs from schematic using Traxmaker PRO and systematically troubleshot circuits with multimeter to ensure full functionality. Programmed PIC16f18875 microcontroller with autonomous logic in PICBasic Pro to identify flames and activate suppression. Developed pathfinding routines using ultrasonic/IR sensors for 100% consistent flame detection.

Triboelectric Nanogenerator (TENG) — *SolidWorks, 3D Modeling, Prototyping*
Designed and modeled a vortex-whistle TENG in SolidWorks to convert wind energy into electricity for portable lighting. Prototyped via 3D printing, optimizing materials for triboelectric output (0.4V measured). Collaborated in a team to iterate design and prepare technical documentation.

First Person Shooter Game *Python, Ursina Game Engine, 3D Assests*

Built core player controls and shooting mechanics in Python while importing 3D models from Sketchfab into the game. Fixed asset loading errors, texture issues, and model rotation problems by debugging file paths and testing different settings. Created a solid starting point for the game by building out the level with simple shapes, ready to add more features and detailed models later.

Distribution Fitting Web App *Python, Streamlit, NumPy, Pandas, Matplotlib, SciPy*

Created a web app that lets users type in numbers or upload CSV files, then automatically generates summary statistics and histogram plots. Implemented distribution fitting for 10 different statistical models (normal, gamma, exponential, etc.) using SciPy's maximum likelihood estimation. Added interactive features including manual parameter adjustment with real-time plot updates and one-click download of analysis results

Simulated Robot *NVIDIA Isaac Sim, ROS2, Python, RViz*

Assembled a simple wheeled robot model in Isaac Sim by adding wheels, chassis, and sensors to create a basic simulated robot. Learned to apply physics properties and joints to make the wheels move and respond to commands in the simulation environment. Set up keyboard controls to drive the robot around and practiced navigating the Isaac Sim interface and workflow. Installed and configured ROS2 Humble with Isaac Sim, will be creating ActionGraphs to stream RGB and lidar data, and visualizing live sensor data in RViz for real-time monitoring

Portfolio Website *HTML, CSS*

Designed and built a responsive personal website using HTML and CSS to display my resume and projects. Structured the site with multiple pages including home, work, hardware, and software sections for easy navigation. Styled the layout with custom fonts, colors, and spacing to create a clean, professional look. Learned core web development fundamentals including flexbox, responsive design, and version control with Git/GitHub

Education

University of Waterloo Expected 2030

B.A.Sc. Honours Nanotechnology Engineering

- Awarded the University of Waterloo President's Scholarship for outstanding academic performance
- Extracurricular: Waterloo Rocketry Team